1 The basic operators

1.1 The killed walk transition operator

Let $P = P_V$ be the killed simple random walk transition operator on V, defined by

$$(Pf)(x) := \frac{1}{2d} \sum_{\substack{y \sim x \\ y \in V}} f(y), \qquad x \in V.$$

Equivalently, its matrix elements are

$$P(x,y) = \frac{1}{2d} \mathbf{1}_{\{x \sim y,\ x,y \in V\}}.$$

This is the transition kernel of simple random walk before exiting V.

Define

$$L := I - P.$$

This is the normalized Dirichlet Laplacian (or killed Laplacian) on V.

1.2 The discrete Dirichlet form

Define, for $f, g : \mathbb{Z}^d \to \mathbb{R}$ supported in V,

$$\mathcal{E}(f,g) := \frac{1}{4d} \sum_{\{x,y\}} (f(x) - f(y))(g(x) - g(y)),$$

where the sum is over unordered nearest-neighbor edges of $\mathbb{Z}^d$.

In particular,

$$\mathcal{E}(f,f) = \frac{1}{4d} \sum_{\{x,y\}} (f(x) - f(y))^2.$$

It is convenient to introduce the inner product

$$\langle f, g \rangle_{\nabla} := 2\mathcal{E}(f,g).$$

Explicitly,

$$\langle f, g \rangle_{\nabla} = \frac{1}{2d} \sum_{\{x,y\}} (f(x) - f(y))(g(x) - g(y)).$$

Proposition 1 (Dirichlet form and operator identity)**.** *For all functions f, g supported in V,*

$$\langle f, g \rangle_{\nabla} = \sum_{x \in V} f(x)\,(Lg)(x).$$

Equivalently,

$$2\mathcal{E}(f,g) = (f, Lg)_{\ell^2(V)}.$$

Proof. Starting from the definition,

$$\sum_{\{x,y\}} (f(x) - f(y))(g(x) - g(y)).$$

For each edge $\{x,y\}$,

$$(f(x) - f(y))(g(x) - g(y)) = f(x)(g(x) - g(y)) + f(y)(g(y) - g(x)).$$

Summing over all unordered edges and then regrouping by the vertex carrying f, we get

$$\sum_{\{x,y\}} (f(x) - f(y))(g(x) - g(y)) = \sum_{x \in V} f(x) \sum_{y \sim x} (g(x) - g(y)).$$

Since $g(y) = 0$ for $y \notin V$, this becomes

$$\sum_{\{x,y\}} (f(x) - f(y))(g(x) - g(y)) = \sum_{x \in V} f(x) \Big(2d\, g(x) - \sum_{\substack{y \sim x \\ y \in V}} g(y) \Big).$$

Divide by $2d$:

$$\langle f, g \rangle_\nabla = \sum_{x \in V} f(x) \Big(g(x) - \frac{1}{2d} \sum_{\substack{y \sim x \\ y \in V}} g(y) \Big) = \sum_{x \in V} f(x)\, (Lg)(x).$$

This is the desired identity. □

Corollary 2. *For all f,*

$$\mathcal{E}(f, f) = \frac{1}{2} (f, Lf)_{\ell^2(V)}.$$

Proposition 3. *L is symmetric and strictly positive definite on $\mathbb{R}^V$.*

Proof. Symmetry follows from the symmetry of $P(x, y)$.

For positivity,

$$(f, Lf)_{\ell^2(V)} = \langle f, f \rangle_\nabla = \frac{1}{2d} \sum_{\{x,y\}} (f(x) - f(y))^2 \geq 0.$$

If $(f, Lf) = 0$, then $f(x) = f(y)$ on every edge. Hence f is constant on every connected component of the graph induced by V together with its boundary edges. Since $f = 0$ outside V, each component touching the boundary must have constant value 0. Therefore $f \equiv 0$ on V. So L is strictly positive definite. □

2 The Green function as the inverse kernel

Definition 4 (Green function)**.** The Green function of the killed walk on V is

$$G(x, y) := \mathbb{E}_x \Big[\sum_{n=0}^{\tau_{V^c} - 1} \mathbf{1}_{\{X_n = y\}} \Big], \qquad x, y \in V.$$

Since the walk is killed upon exiting V, the n-step killed transition kernel is $P^n(x, y)$. Therefore

$$G(x, y) = \sum_{n=0}^{\infty} P^n(x, y).$$

Proposition 5 (Green operator identity)**.** *As matrices on $\mathbb{R}^V$,*

$$G = \sum_{n=0}^{\infty} P^n = (I - P)^{-1} = L^{-1}.$$

Equivalently,

$$LG = GL = I.$$

Proof. Because V is finite and the walk is killed upon exiting V, eventually the walk leaves V almost surely, so the Neumann series converges entrywise. Then

$$(I-P)\sum_{n=0}^{\infty} P^n = \sum_{n=0}^{\infty} P^n - \sum_{n=1}^{\infty} P^n = I.$$

Hence

$$L\,G = (I-P)G = I.$$

Similarly $G(I-P) = I$. Thus $G = L^{-1}$. $\square$

Remark 6. This is the precise relation between the Dirichlet form and the Green function:

$$\textit{Dirichlet form} \longleftrightarrow \textit{operator } L \quad \textit{and} \quad \textit{Green function} = L^{-1}.$$

So the Green function is *not* the Dirichlet form itself; it is the inverse kernel of the operator represented by that form.

3 Construction (A): Gibbs density

Definition 7 (Gibbs construction)**.** The zero-boundary DGFF on V may be defined by the density

$$\mu_V(\mathrm{d}h) = \frac{1}{Z_V} \exp\Big\{-\frac{1}{4d}\sum_{\{x,y\}} (h(x)-h(y))^2\Big\} \prod_{x\in V} \mathrm{d}h(x),$$

where $h(x) = 0$ for $x \notin V$.

Using the Dirichlet form,

$$\mu_V(\mathrm{d}h) = \frac{1}{Z_V} \exp\{-\mathcal{E}(h,h)\} \prod_{x\in V} \mathrm{d}h(x).$$

By the identity proved above,

$$\mathcal{E}(h,h) = \frac{1}{2}(h, Lh)_{\ell^2(V)}.$$

Hence

$$\mu_V(\mathrm{d}h) = \frac{1}{Z_V} \exp\Big\{-\frac{1}{2}(h, Lh)_{\ell^2(V)}\Big\} \prod_{x\in V} \mathrm{d}h(x).$$

Proposition 8. *Construction (A) defines a centered Gaussian law on $\mathbb{R}^V$ with precision matrix L and covariance matrix L^{-1}.*

Proof. Since L is symmetric positive definite, it admits an orthonormal diagonalization

$$L = U^\top D U,$$

where U is orthogonal and $D = \mathrm{diag}(\lambda_1, \dots, \lambda_{|V|})$ with all $\lambda_i > 0$.

Let

$$z := D^{1/2} U h.$$

Then

$$(h, Lh) = h^\top U^\top D U h = z^\top z = |z|^2.$$

The Jacobian is constant, so the density becomes

$$\propto \exp\Big\{-\frac{1}{2}|z|^2\Big\}\,\mathrm{d}z,$$

which is the standard Gaussian density in the z-coordinates. Thus h is a centered Gaussian vector with covariance

$$U^\top D^{-1} U = L^{-1}.$$

$\square$

4 Construction (B): covariance given by the Green function

Definition 9 (Green covariance construction)**.** Define a centered Gaussian field $(\phi_x)_{x\in V}$ by

$$\mathbb{E}(\phi_x) = 0, \qquad \mathbb{E}(\phi_x\phi_y) = G(x,y).$$

This construction is well-defined because $G = L^{-1}$ is symmetric positive definite.

Theorem 10 (Equivalence of **(A)** and **(B)**)**.** *The Gibbs construction **(A)** and the Green covariance construction **(B)** define the same law on $\mathbb{R}^V$.*

Proof. By the previous proposition, the Gibbs construction gives a centered Gaussian field with covariance matrix L^{-1}.

By the Green operator identity,

$$L^{-1} = G.$$

Therefore the covariance matrix of the Gibbs field is exactly G. Since centered Gaussian laws are completely determined by their covariance matrices, the two constructions coincide. $\square$

5 Construction (C): Hilbert-space Gaussian series

5.1 The Hilbert space

Let

$$\mathcal{H}_V := \left\{ f : \mathbb{Z}^d \to \mathbb{R} : \mathrm{supp}(f) \subset V \right\}.$$

Since V is finite, $\mathcal{H}_V$ is a finite-dimensional Hilbert space with inner product

$$\langle f, g\rangle_\nabla := 2\mathcal{E}(f,g) = \sum_{x\in V} f(x)(Lg)(x).$$

Choose an orthonormal basis $\{\psi_1, \ldots, \psi_{|V|}\}$ of $\mathcal{H}_V$ with respect to $\langle\cdot,\cdot\rangle_\nabla$. Let $Z_1, \ldots, Z_{|V|}$ be i.i.d. standard normal random variables. Define

$$\eta(x) := \sum_{k=1}^{|V|} Z_k\,\psi_k(x), \qquad x \in V.$$

Proposition 11. *$\eta = (\eta(x))_{x\in V}$ is a centered Gaussian field, and*

$$\mathbb{E}[\eta(x)\eta(y)] = \sum_{k=1}^{|V|} \psi_k(x)\psi_k(y).$$

Proof. Each $\eta(x)$ is a linear combination of independent Gaussian random variables, so the vector η is Gaussian. It is centered because $\mathbb{E}Z_k = 0$.

Moreover,

$$\mathbb{E}[\eta(x)\eta(y)] = \sum_{k,\ell} \mathbb{E}[Z_k Z_\ell]\psi_k(x)\psi_\ell(y) = \sum_{k=1}^{|V|} \psi_k(x)\psi_k(y),$$

since $\mathbb{E}[Z_k Z_\ell] = \delta_{k\ell}$. $\square$

5.2 Riesz representation and the Green kernel

We now show that the covariance above is exactly the Green function.

Proposition 12 (Green function as reproducing kernel)**.** *For each fixed $x \in V$, the function $G_x(\cdot) := G(x, \cdot)$ belongs to $\mathcal{H}_V$ and satisfies*

$$\langle G_x, f \rangle_\nabla = f(x) \qquad \textit{for all } f \in \mathcal{H}_V.$$

Proof. Since $G = L^{-1}$, we have

$$LG_x = \delta_x,$$

where $\delta_x(y) = \mathbf{1}_{\{x=y\}}$.

Using the identity $\langle u, v \rangle_\nabla = (u, Lv)_{\ell^2(V)}$ and the symmetry of L,

$$\langle G_x, f \rangle_\nabla = (G_x, Lf)_{\ell^2(V)} = (LG_x, f)_{\ell^2(V)} = (\delta_x, f)_{\ell^2(V)} = f(x).$$

□

Corollary 13. *For each fixed $x \in V$,*

$$G_x = \sum_{k=1}^{|V|} \langle G_x, \psi_k \rangle_\nabla \ \psi_k = \sum_{k=1}^{|V|} \psi_k(x)\psi_k.$$

Therefore,

$$G(x, y) = \sum_{k=1}^{|V|} \psi_k(x)\psi_k(y).$$

Proof. Expand G_x in the orthonormal basis $\{\psi_k\}$:

$$G_x = \sum_{k=1}^{|V|} \langle G_x, \psi_k \rangle_\nabla \ \psi_k.$$

By the reproducing identity,

$$\langle G_x, \psi_k \rangle_\nabla = \psi_k(x).$$

Thus

$$G_x = \sum_{k=1}^{|V|} \psi_k(x)\psi_k.$$

Evaluating at y yields

$$G(x, y) = \sum_{k=1}^{|V|} \psi_k(x)\psi_k(y).$$

□

Theorem 14 (Equivalence of **(B)** and **(C)**)**.** *The Green covariance construction **(B)** and the Hilbert-space Gaussian series construction **(C)** define the same law on $\mathbb{R}^V$.*

Proof. The field η from construction **(C)** is centered Gaussian, and its covariance is

$$\mathbb{E}[\eta(x)\eta(y)] = \sum_{k=1}^{|V|} \psi_k(x)\psi_k(y) = G(x, y).$$

Thus η is a centered Gaussian field with covariance G, exactly as in construction **(B)**. Hence the laws coincide. □

6 Final equivalence theorem

Theorem 15 (Equivalence of the three constructions)**.** *For a finite set $V \subset \mathbb{Z}^d$ with zero boundary condition, the following define the same law on $\mathbb{R}^V$:*

(A) *the Gibbs density*

$$\mu_V(\mathrm{d}h) \propto \exp\Big\{-\frac{1}{4d}\sum_{\{x,y\}}(h(x)-h(y))^2\Big\}\prod_{x\in V}\mathrm{d}h(x),$$

(B) *the centered Gaussian field with covariance*

$$\mathbb{E}[h(x)h(y)] = G(x,y),$$

where G is the Green function of the killed random walk on V,

(C) *the Hilbert-space Gaussian series*

$$h(x) = \sum_{k=1}^{|V|} Z_k\,\psi_k(x),$$

where $\{\psi_k\}$ is any orthonormal basis of $(\mathcal{H}_V, \langle\cdot,\cdot\rangle_\nabla)$.

Proof. We have proved:

$$\textbf{(A)} = \textbf{(B)} \qquad \text{and} \qquad \textbf{(B)} = \textbf{(C)}.$$

Therefore all three constructions are equivalent. □

7 A concise conceptual summary

The correct logical chain is

$$\boxed{\text{Dirichlet form } \mathcal{E}} \iff \boxed{\text{operator } L = I - P} \iff \boxed{\text{Green kernel } G = L^{-1}}.$$

Then:

- the Gibbs construction uses the quadratic form $\mathcal{E}(h,h)$;
- this quadratic form is $\frac{1}{2}(h, Lh)$;
- hence the Gibbs law is Gaussian with precision L and covariance L^{-1};
- but $L^{-1} = G$, the Green function;
- and in the Hilbert-space picture, $G(x,\cdot)$ is the Riesz representer of the evaluation map $f \mapsto f(x)$, which yields the Gaussian series construction.

So the three constructions are not merely analogous; they are exactly the same Gaussian field written in three different languages.